Scientists Find Missing Matter Hiding Around Galaxies

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Scientists have long figured out how much matter should be in the **universe**. But more than half of the ordinary matter seemed to be missing. Now they think they know where some of it is hiding.

Ordinary matter is made of tiny particles, like the **protons** and neutrons in stars, planets and people. It is different from **dark matter**, a mystery stuff that does not give off light.

Haochen Wang, a student at the Massachusetts Institute of Technology, led the new study. He says the missing matter is a thin gas spread out in huge, faint clouds around **galaxies**. In some places, there is less than one tiny particle in a space the size of a washing machine!

To find it, the team used fast radio bursts. These are quick flashes of **radio waves** from far away. When a burst passes through gas, its signal gets stretched out. More gas means more stretching.

Not everyone agrees this is the first discovery. But **astronomer** Vikram Ravi called it "a great signpost of what is to come."

Word Treasure

universe -- Everything that exists, including all of space, stars and galaxies.
galaxies -- Huge groups of billions of stars held together by gravity.
protons -- Tiny particles found inside every atom.
dark matter -- An invisible kind of stuff in space that does not give off light.
radio waves -- Invisible waves of energy that can travel very far through space.
astronomer -- A scientist who studies stars, planets and space.

Background you need

Everything around us, from people to stars, is made of tiny particles.

Some things in space are too faint or spread out for telescopes to see.

Scientists can learn about hidden things by watching how signals change as they travel through space.

Break it down

WHO: Haochen Wang and his team at the Massachusetts Institute of Technology
WHAT: They found missing ordinary matter in thin gas clouds
WHERE: Around galaxies far out in space
WHY: To solve the puzzle of where more than half of the ordinary matter went

Why it matters

Finding missing matter helps us understand what everything in the universe, including you, is made of.

Test what you remember

1. What did scientists think was missing?

- (A) More than half of the ordinary matter
- (B) The Moon
- (C) Earth's oceans
- (D) All the stars

2. Where is some of the missing matter hiding?

- (A) In thin gas clouds around galaxies
- (B) Inside volcanoes
- (C) On the Moon
- (D) Under the sea

3. Who led the new study?

- (A) Carson Bruns
- (B) Haochen Wang
- (C) Vikram Ravi
- (D) Asaph Hall

4. What did the team use to find the matter?

- (A) Magnets
- (B) Fast radio bursts
- (C) Rockets
- (D) Microscopes

5. What happens to a burst's signal when it passes through gas?

- (A) It turns blue
- (B) It gets stretched out
- (C) It disappears
- (D) It gets louder

6. How much space has less than one tiny particle in some places?

- (A) A space the size of a washing machine
- (B) A space the size of a pea
- (C) A space the size of a city
- (D) A space the size of Earth

Answer key (try the quiz first!): 1: A 2: A 3: B 4: B 5: B 6: A

Questions to think about

1. Haochen Wang: Says the missing matter is a thin gas spread out in huge, faint clouds.

2. Vikram Ravi: Not sure it is the first discovery, but calls it "a great signpost of what is to come."

MY THOUGHTS (at least 20 words):